Week 13 Firewall

06016421 IT Infrastructure Security
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(Reference : https://csrc.nist.gov/pubs/sp/800/41/r1/final)

Firewall - Descriptions and Roles (P.7/48)

- Firewalls are devices or programs that control the flow of network traffic between networks or hosts that employ differing security postures.
- At one time, most firewalls were deployed at network perimeters.
  - This provided some measure of protection for internal hosts, but it could not recognize all instances and forms of attack, and attacks sent from one internal host to another often do not pass through network firewalls.
  - Because of these and other factors, network designers now often include firewall functionality at places other than the network perimeter to provide an additional layer of security, as well as to protect mobile devices that are placed directly onto external networks.
- Threats have gradually moved from being most prevalent in lower layers of network traffic to the application layer, which has reduced the general effectiveness of firewalls in stopping threats carried through network communications.
  - However, firewalls are still needed to stop the significant threats that continue to work at lower layers of network traffic. Firewalls can also provide some protection at the application layer, supplementing the capabilities of other network security technologies.
- There are several types of firewalls, each with varying capabilities to analyze network traffic and allow or block specific instances by comparing traffic characteristics to existing policies.
  - Understanding the capabilities of each type of firewall, and designing firewall policies and acquiring firewall technologies that effectively address an organization's needs, are critical to achieving protection for network traffic flows.

Guidelines for Firewall Implementation (P.7/48)

To improve the effectiveness and security of their firewalls, organizations should implement the following recommendations:

- **Create a firewall policy** that specifies how firewalls should handle inbound and outbound network traffic.
  - A firewall policy defines how an organization's firewalls should handle inbound and outbound network traffic for specific IP addresses and address ranges, protocols, applications, and content types based on the organization's information security policies.
  - Organizations should conduct risk analysis to develop a list of the types of traffic needed by the organization and how they must be secured—including which types of traffic can traverse a firewall under what circumstances.
- **Identify all requirements** that should be considered when determining which firewall to implement.
  - Organizations need to determine which network areas need to be protected, and which types of firewall technologies will be most effective for the types of traffic that require protection.
  - There are many considerations that organizations should include in their firewall selection and planning processes.
- **Create rulesets** that implement the organization's firewall policy while supporting firewall performance.
  - Firewall rulesets should be as specific as possible with regards to the network traffic they control.
  - To create a ruleset involves determining what types of traffic are required, including protocols the firewall may need to use for management purposes.
  - Details of creating rulesets vary by type of firewall and specific products, but many firewalls can have their performance improved by optimizing firewall rulesets.
- **Manage** firewall architectures, policies, software, and other components throughout the life of the firewall solutions.
  - There are many aspects to firewall management.
    - For example, choosing the type or types of firewalls to deploy and their positions within the network can significantly affect the security policies that the firewalls can enforce.
  - Policy rules may need to be updated as the organization's requirements change, such as when new applications or hosts are implemented within the network.
  - Firewall rulesets and policies should be managed by a formal change management control process because of their potential to impact security and business operations, with ruleset reviews or tests performed periodically to ensure continued compliance with the organization's policies.
  - Logs and alerts should be continuously monitored to identify threats—both successful and unsuccessful.
  - Firewall performance needs to be monitored to enable potential resource issues to be identified and addressed before components become overwhelmed.
  - Firewall software should be patched as vendors provide updates to address vulnerabilities.

Overview of Firewall Technologies (P.11/48)

- While firewalls are often discussed in the context of Internet connectivity, they may also have applicability in other network environments. For example, many enterprise networks employ firewalls to restrict connectivity to and from the internal networks used to service more sensitive functions, such as accounting or personnel.
- By employing firewalls to control connectivity to these areas, an organization can prevent unauthorized access to its systems and resources. Inclusion of a proper firewall provides an additional layer of security.
- Organizations often need to use firewalls to meet security requirements from mandates (e.g., FISMA); some mandates, such as the Payment Card Industry (PCI) Data Security Standard (DSS), specifically require firewalling.

Application Layer. This layer sends and receives data for particular applications, such as Domain Name System (DNS), Hypertext Transfer Protocol (HTTP), and Simple Mail Transfer Protocol (SMTP). The application layer itself has layers of protocols within it. For example, SMTP encapsulates the Request for Comments (RFC) 2822 message syntax, which encapsulates Multipurpose Internet Mail Extensions (MIME), which can encapsulate other formats such as Hypertext Markup Language (HTML).
Transport Layer. This layer provides connection-oriented or connectionless services for transporting application layer services between networks, and can optionally ensure communications reliability. Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are commonly used transport layer protocols.
IP Layer (also known as the Network Layer). This layer routes packets across networks. Internet Protocol version 4 (IPv4) is the fundamental network layer protocol for TCP/IP. Other commonly used protocols at the network layer are Internet Protocol version 6 (IPv6), ICMP, and Internet Group Management Protocol (IGMP).
Hardware Layer (also known as the Data Link Layer). This layer handles communications on the physical network components. The best known data link layer protocol is Ethernet.

Firewall Technologies (P.12/48)

- Packet Filtering : (a.k.a. stateless inspection firewalls)
- Stateful Inspection : Track the state of packets that deviate from the expected state.

Table 2-1. State Table Example

Source Address	Source Port	Destination Address	Destination Port	Connection State
192.168.1.100	1030	192.0.2.71	80	Initiated
192.168.1.102	1031	10.12.18.74	80	Established
192.168.1.101	1033	10.66.32.122	25	Established
192.168.1.106	1035	10.231.32.12	79	Established

- Application Firewalls
  - Allow or deny access based on how an application is running over the network.
    - For instance, an application firewall can determine if an email message contains a type of attachment that the organization does not permit (such as an executable file), or if instant messaging (IM) is being used over port 80 (typically used for HTTP).
    - Another feature is that it can block connections over which specific actions are being performed (e.g., users could be prevented from using the FTP "put" command, which allows users to write files to the FTP server).
    - Allow or deny web pages that contain particular types of active content, such as Java or ActiveX, or that have SSL certificates signed by a particular certificate authority (CA), such as a compromised or revoked CA.
- Application-Proxy Gateways
  - Combines lower-layer access control with upper-layer functionality.
  - Contain a proxy agent that acts as an (trans hosts that wish to communicate with each other, and never allows a direct connection between them.
- Dedicated Proxy Servers
  - Retain proxy control of traffic
  - Have much more limited firewalling capabilities.
- Virtual Private Networking (VPN)
- Network Access Control
  - Perform client checks for incoming connections from remote users and allow or disallow access based on those checks. This checking, commonly called network access control (NAC) or network access protection (NAP), allows access based on the user's credentials and the results of performing "health checks" on the user's computer. Health checks typically consist of verifying that one or more of the following comply with organizational policy:
    - Latest updates to antimalware and personal firewall software
    - Configuration settings for antimalware and personal firewall software
    - Elapsed time since the previous malware scan
    - Patch level of the operating system and selected applications
    - Security configuration of the operating system and selected applications
- Unified Threat Management (UTM) : Many firewalls combine multiple features into a single system.
- Web Application Firewalls (WAF) : Put in front of web servers to prevent attacks on the server.
- Firewalls for Virtual Infrastructures : Monitor virtualized machines and networks.

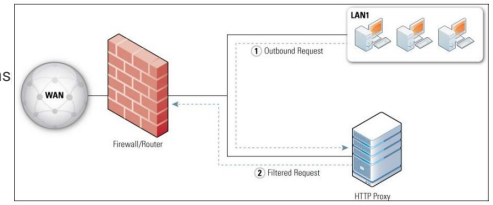


Figure 2-2. Application Proxy Configuration

Firewalls for Individual Hosts and Home Networks (P.21/48)

Host-Based Firewalls and Personal Firewalls

Personal Firewall Appliances

Limitations of Firewall Inspection (P.21/48)

Firewalls can only work effectively on traffic that they can inspect.

- A firewall that cannot understand the traffic flowing through it will not handle that traffic properly—for example, allowing traffic that should be blocked.
  - Encrypted Traffic
  - Tunneled Traffic
- Firewall's rules will determine what to do with traffic it does not (or, in the case of encrypted traffic, cannot) understand.
  - An organization should have policies about how to handle traffic in such cases, such as either permitting or blocking encrypted traffic that is not authorized to be encrypted.
    - จะให้ทำยังไงกับ Traffic ที่ถูก Encrypted หรือ Tunneled โดยไม่ได้รับอนุญาต

Recommendations for Firewall Implementation (P.22/48)

- The use of NAT should be considered a form of routing, not a type of firewall.
- Organizations should only permit outbound traffic that uses the source IP addresses in use by the organization.
  - คำถาม 01 : เพื่อป้องกัน Threat ใดตามนิยาม STRIDE Threat Model ?
- Compliance checking is only useful in a firewall when it can block communication that can be harmful to protected systems.
- When choosing the type of firewall to deploy, it is important to decide whether the firewall needs to act as an application proxy.
- Management of personal firewalls should be centralized to help efficiently create, distribute, and enforce policies for all users and groups.
  - คำถาม 02 : MS Windows มีเครื่องมืออะไรที่สามารถใช้เพื่อให้เกิดผลตามนี้ได้

Firewalls and Network Architectures (P.23/48)

- Firewalls are used to separate networks with differing security requirements,
  - e.g., the Internet and an internal network that houses servers with sensitive data.
- Organizations should use firewalls wherever their internal networks and systems interface with external networks and systems, and where security requirements vary among their internal networks.
- Organizations should properly determine where firewalls should be placed, and where other networks and systems should be located in relation to the firewalls.
- Network Layouts with Firewalls
  - Simple Routed Network with Firewall Device (Fig. 3-1)
  - Firewall with a DMZ (Fig. 3-2)
- Firewalls Acting as Network Address Translators
  - Typically, a NAT acts as a router that has a network with private addresses on the inside and a single public address on the outside. The way a NAT performs this many-to-one mapping varies between implementations, but almost always involves the following:
  - Hosts on the inside network initiating connections to the outside network cause the NAT to map the source port of the connection to a different source port that is controlled by the NAT. The NAT uses this source port number to map connections from the outside back to the host on the inside.
  - Hosts on the outside of the network cannot initiate contact with hosts on the inside network. In some firewalls, the NAT can be configured to map a particular destination port on the NAT to a particular host on the inside of the NAT; for example, all HTTP requests that go to the NAT could be directed to a single host on the protected side of the firewall. This feature is sometimes called pinholing.
- Architecture with Multiple Layers of Firewalls
  - No limitation on where a firewall can be placed in a network. While firewalls should be at the edge of a logical network boundary, creating an “inside” and “outside” on either side of the firewall, a network administrator may wish to have additional boundaries within the network and deploy additional firewalls to establish such boundaries.
  - The use of multiple layers of firewalls is quite common to provide defense-in-depth.

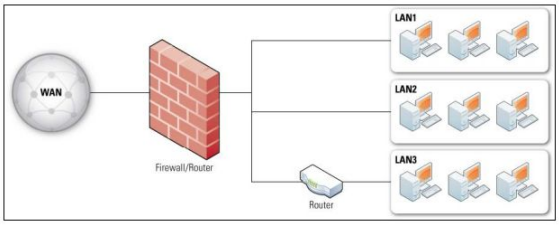


Figure 3-1. Simple Routed Network with Firewall Device

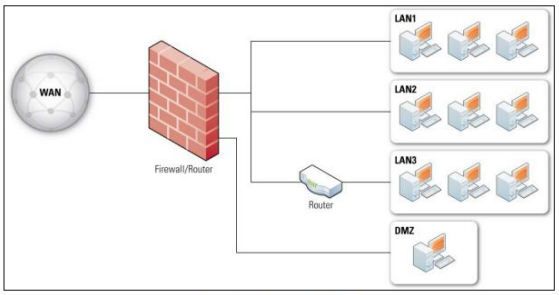


Figure 3-2. Firewall with a DMZ

Recommendations for Firewalls and Network Architectures (P.26/48)

- In general, a firewall should fit into a current network’s layout. However, an organization might change its network architecture at the same time as it deploys a firewall as part of an overall security upgrade.
- Different common network architectures lead to very different choices for where to place a firewall, so an organization should assess which architecture works best for its security goals.
- If an edge firewall has a DMZ, consider which outward-facing services should be run from the DMZ and which should remain on the inside network.
- Do not rely on NATs to provide the benefits of firewalls.
- In some environments, putting one firewall behind another may lead to a desired security goal, but in general such multiple layers of firewalls can be troublesome. (For further details, read Topic 3.3 P.26/48)

Firewall Policy (P.28/48)

- A firewall policy dictates how firewalls should handle network traffic for specific IP addresses and address ranges, protocols, applications, and content types (e.g., active content) based on the organization's information security policies.
    - Before a firewall policy is created, some form of risk analysis should be performed to develop a list of the types of traffic needed by the organization and categorize how they must be secured — including which types of traffic can traverse a firewall under what circumstances.
    - This risk analysis should be based on an evaluation of threats; vulnerabilities; countermeasures in place to mitigate vulnerabilities; and the impact if systems or data are compromised.
    - Firewall policy should be documented in the system security plan and maintained and updated frequently as classes of new attacks or vulnerabilities arise, or as the organization's needs regarding network applications change.
    - The policy should also include specific guidance on how to address changes to the ruleset.
- Generally, firewalls should block all inbound and outbound traffic that has not been expressly permitted by the firewall policy—traffic that is not needed by the organization.
    - This practice, known as deny by default, decreases the risk of attack and can also reduce the volume of traffic carried on the organization's networks.
    - Because of the dynamic nature of hosts, networks, protocols, and applications, deny by default is a more secure approach than permitting all traffic that is not explicitly forbidden.
- Policies Based on IP Addresses and Protocols
    - IP Addresses and Other IP Characteristics; IPv6; TCP and UDP; ICMP; IPsec Protocols
- Policies Based on Applications
    - Inbound application firewalls or application proxies take a different approach—they let traffic destined for a particular server into the network, but capture that traffic in a server that processes it like a port-based firewall.
    - The application-based approach provides an additional layer of security for incoming traffic by validating some of the traffic before it reaches the desired server. The theory is that the inbound application firewall's or proxy's additional security layer can protect the server better than the server can protect itself—and can also remove malicious traffic before it reaches the server to help reduce server load.
- Policies Based on User Identity : One of the most common ways to enforce user identity policy at a firewall is by using a VPN.
- Policies Based on Network Activity : ใช้เงื่อนไขเช่น Session Idle Period, Incoming Traffic/Request Rate, Source Attributes,...

Recommendations for Firewall Policy (P.34/48)

- An organization's firewall policy should be based on a comprehensive risk analysis.
    - คำถาม 01 : Risk analysis มีอะไรบ้าง
- Firewall policies should be based on blocking all inbound and outbound traffic, with exceptions made for desired traffic.
    - คำถาม 02 : เทียบได้กับ Blacklist policy หรือ Whitelist policy
- Policies should take into account the source and destination of the traffic in addition to the content.
- Many types of IPv4 traffic, such as that with invalid or private addresses, should be blocked by default.
- Organizations should have policies for handling incoming and outgoing IPv6 traffic.
- An organization should determine which applications may send traffic into or out of its network and make firewall policies to block traffic for other applications.

Firewall Planning and Implementation (P.35/48)

A successful firewall deployment can be achieved by following a clear, step-by-step planning and implementation process.

The use of a phased approach for deployment can minimize unforeseen issues and identify potential pitfalls early on.

- **Plan**. The first phase of the process involves identifying all requirements that an organization should consider when determining which firewall to implement to enforce the organization's security policy.
- **Configure**. The second phase involves all facets of configuring the firewall platform. This includes installing hardware and software as well as setting up rules for the system.
- **Test**. The next phase involves implementing and testing a prototype of the designed solution in a lab or test environment. The primary goals of testing are to evaluate the functionality, performance, scalability, and security of the solution, and to identify any issues—such as interoperability—with components.
- **Deploy**. Once testing is completed and all issues are resolved, the next phase focuses on deployment of the firewall into the enterprise.
- **Manage**. After the firewall has been deployed, it is managed throughout its lifecycle to include component maintenance and support for operational issues. This lifecycle process is repeated when enhancements or significant changes need to be incorporated into the solution.

Firewall Planning and Implementation - Plan (P.35/48)

Basic principles that organizations should follow in the planning of firewall deployments

- Use devices as they were intended to be used
  - E.g., routers are meant to handle routing, not highly complex filtering, which can cause an excess burden on the router's processor.
  - Additionally, firewalls should not be expected to provide non-security services, such as acting as a web server or email server.
- Create defense-in-depth
  - Defense-in-depth can be accomplished by using multiple firewalls throughout an organization, including at the perimeter, in front of sensitive internal departments, and on individual computers.
- Pay attention to internal threats
  - Focusing attention solely on external threats leaves the network wide open to attacks from within. These threats may not come directly from insiders, but can involve internal hosts infected by malware or otherwise compromised by external attackers. Important internal systems should be placed behind internal firewalls.
- Document the firewall's capabilities
  - Each model of firewall has different capabilities and limitations. These will sometimes affect the planning of the organization's security policy and firewall deployment strategy. Any features that positively or negatively affect this planning should be written into the overall planning document.

Organizations should consider the following when purchasing and implementing a firewall solution

- Security Capabilities : Features, Supported Technologies or Protocols, Network areas to be protected, …
- Management : How to manage the firewall?
- Performance (generally for network firewalls only)
- Integration : Compatibility, Interoperability, Additional hardware/software requirements
- Physical Environment : พื้นที่ติดตั้ง, ระบบไฟ, ระบบปรับอากาศ, ระบบป้องกันภัยต่างๆเช่นอัคคีภัย ฯลฯ
  (generally a consideration for network firewalls, although it may also apply to the centralized components of host-based firewall implementations)
- Personnel : Required skill sets of person in charge of firewall admin.
- Future Needs including scalability issue and so on…

Firewall Planning and Implementation - Configure (P.38/48)

The configuration phase involves all facets of configuring the firewall platform. This includes installing hardware and software, configuring policies, configuring logging and alerting, and integrating the firewall into the network architecture.

- Hardware and Software Installation
- Policy Configuration
  - At minimum, the following rules should be defined:
    - Port filtering should be enabled at the outer edge of the network, and probably at places inside the network as well.
    - Content filtering should be done as close to the content receiver as possible.
- Logging and Alerts Configuration
  - Notifications may include the following:
    - Any modifications or disabling of the firewall rules
    - System reboots, disk shortages, and other operational events
    - Secondary system status changes, if applicable

Firewall Planning and Implementation - Test (P.40/48)

- New firewalls should be tested and evaluated before deployment to ensure that they are working properly.
- Testing should be completed on a test network without connectivity to the production network.
- This test network should attempt to replicate the production network as faithfully as possible, including the network topology and network traffic that would travel through the firewall. Aspects of the solution to evaluate include the following:
  - Connectivity
    - Users can establish and maintain connections through the firewall.
  - Ruleset
    - Traffic that is specifically allowed by the security policy is permitted. All traffic that is not allowed by the security policy is blocked.
    - Verification of the ruleset should include both reviewing it manually and testing whether the rules work as expected.
  - Application Compatibility
    - Do not break or interfere with the use of existing software applications.
  - Management
    - Administrators can configure and manage the solution effectively and securely.
  - Logging
    - Logging and data management function in accordance with the organization's policies and strategies.
  - Performance
    - Solutions provide adequate performance during normal and peak usage.
  - Security of the Implementation
    - The firewall implementation itself may contain vulnerabilities and weaknesses that attackers could exploit
  - Component Interoperability
    - Components of the firewall solution must function together properly.
  - Policy Synchronization
    - If there are multiple firewalls running synchronized policies or groups of rules, test that the synchronization works in various scenarios (such as if one or more nodes are offline).
  - Additional Features
    - Additional features that will be used by the firewall—such as VPN and antimalware capabilities—should be tested to ensure they are working properly.

Firewall Planning and Implementation - Deploy (P.40/48)

- Once testing is complete and all issues have been resolved, the next phase of the firewall planning and implementation model is deployment, which should be done in accordance with organization policies.
- Before deploying the firewall,
  - Administrators should notify users or owners of potentially affected systems of the planned deployment, and instruct them who to notify if they encounter any problems.
  - Any changes required to other equipment should also be coordinated as part of the firewall deployment.
  - The security policy expressed by the firewall's configuration should be added to the overall security policy of the organization, and the ongoing changes to its configuration should be integrated with the organization's configuration management processes.
  - If multiple firewalls are being deployed, including personal firewalls or at multiple branch offices,
    - A gradual or phased approach should be considered;
    - A pilot program would also be helpful, especially for identifying and resolving issues of conflicting policies.
  - This will provide administrators with an opportunity to evaluate the firewall solution's impact and resolve issues before enterprise-wide deployment.
- Connecting a firewall to the organization's network requires more than just inserting the firewall into the traffic flow from the outside of the network to the inside: it also involves integrating the firewall with other network elements that will interact with the firewall.

Firewall Planning and Implementation - Manage (P.41/48)

- This last phase of the firewall planning and implementation model is the longest lasting, because managing the solution involves maintaining firewall architecture, policies, software, and other components of the solution chosen to be deployed.
- One example of a typical maintenance action is testing and applying patches to firewall devices.
- Policy rules may need to be updated as new threats are identified and requirements change, such as when new applications or hosts are implemented within the network, and should be reviewed periodically to ensure they remain in compliance with security policy.
- It is important to monitor the performance of firewall components to ensure that potential resource issues are identified and addressed before components become overwhelmed.
- Logs and alerts should be monitored continuously to identify threats—successful and unsuccessful—that are made to the system.
- Another important task is to perform periodic testing to verify that firewall rules are functioning as expected.
- Firewall policies and rulesets should be backed up regularly.
- Some firewalls can store this information in multiple formats, such as a binary format that is used to configure the firewall and a human-readable format that can be read by auditors. If multiple formats are available, backups should be kept in all of them.
- Changes to firewall rulesets/policies impact security and should be managed as part of a formal configuration management process. Many firewalls have auditing of changes as part of their administrative interfaces, but this does not necessarily track policy changes. At a minimum, a log should be kept of all policy decisions and ruleset changes—and this log should be associated with the firewall.
- It is best to review the firewall policy at regular intervals so that such reviews do not only happen during policy or security audits (or, worse, only during emergencies).
- Organizations may want to consider conducting penetration testing to assess the overall security of their network environment. This testing can be used to verify that a firewall ruleset is performing as intended by generating network traffic and monitoring how it is handled by the firewall in comparison with its expected response. Penetration testing should be employed in addition to, rather than instead of, a conventional audit program.

Lab 6 : Firewall

- Lab 6-A ติดตั้งและหัดใช้ Fortigate Firewall (FGT) ver. 6.4.15 — Free Version Time Limit
- Lab 6-B สร้าง Firewall Policy ให้ LAN และ WAN ติดต่อกันได้
- Lab 6-C สร้าง Firewall 3 ขา เชื่อมต่อ LAN, WAN, และ DMZ (Option)

Download :

Official Site : https://support.fortinet.com/Download/VMImages.aspx

GNS3 Market Place : https://www.gns3.com/fortigate/-/join

YouTube How to Download & Install ver. 6.4.5 : https://youtu.be/nqJ7-4tB7jM?si=jMIbiyWitWTxfz_r ← **ดูวิธีติดตั้งที่นี่**

Disk Image ver. 6.4.15 : https://www.dropbox.com/scl/fi/it3lq3eday70zx99hanzg/FGT_VM64_KVM-v6.M-build2095-FORTINET.out.kvm.zip?rlkey=7lbdnka8bqnkqap92ht1gu8bt&dl=0

Lab Manual :

https://docs.fortinet.com/document/fortigate/6.4.15/administration-guide/954635/getting-started

https://docs.fortinet.com/product/fortigate/6.4.15

Lab 6-A : Firewall Setup

1. ติดตั้ง Firewall Appliance ใน GNS3

   - ถ้าเป็น Fortigate (FGT) ให้ใช้ ver. 6.X.X (แนะนำ 6.4.15)
     (ห้ามใช้ ver. 7 เพราะ FGT Eval. Ver. จะเลือก TLS/SSL และ Encryption Method ในการทำ VPN ไม่ได้ ทำให้มีปัญหาในการ Connect ด้วย Client ที่ตัวเลือกบังคับเช่นกัน)

2. ติดตั้ง Network และ Firewall ดังรูป

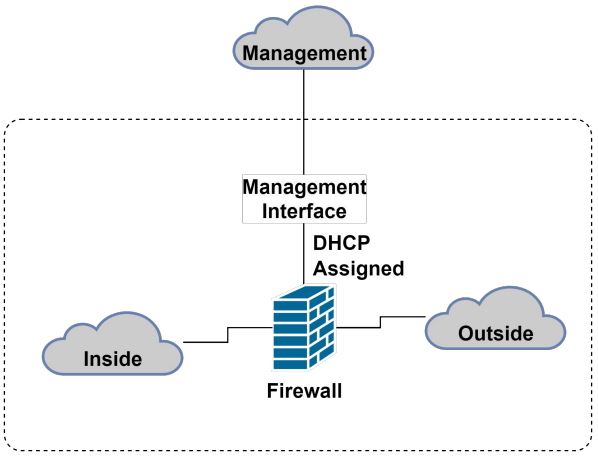
3. ติดตั้ง Management Interface ของ Firewall ให้รับ IP จาก DHCP Server และให้ยอมรับการ PING, SSH และ HTTP

4. ลอง Ping จาก PC ใน Management Network ไปยัง Firewall

5. ลอง Access จาก PC ใน Management Network ไปยัง Firewall ด้วยการ TELNET, SSH และ Browse ผ่าน Web

6. ลองเข้าเมนู Dash Board และเมนูต่างๆที่จะทำให้สามารถคอนฟิก Firewall เช่น Network → Interface, Policy & Objects → Firewall Policy, VPN, User & Authen., Log & Report, และ Config. Management

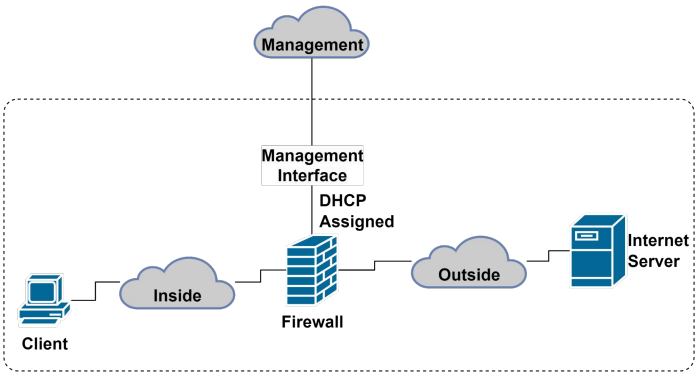
7. บันทึกสิ่งที่ได้ทำและผลการทดลองข้อ 4-6



Zone	IP Address
Management	10.30.6.0/24
Outside	203.0.113.0/24
Inside	192.168.1.0/24

Lab 6-B : สร้าง Firewall Policy ให้ LAN และ WAN ติดต่อกันได้

1. ติดตั้ง Network และ Firewall ดังรูป โดยห้ามใช้ NAT (อย่าลืมปิด NAT)
2. ติดตั้ง Web Server, Telnet Server SSH Server บน Internet Server
3. คอนฟิก Firewall ให้ Client และ Internet Server สามารถ Ping, Browse WEB และ SSH ไปมาหากันได้ แล้วบันทึกสิ่งที่ทำและผลที่ได้
4. ลองคอนฟิก Firewall Policy ให้ Block สิ่งที่ทำในข้อ 3 ได้ทั้งหมด ยกเว้นให้ Client สามารถ Ping ไปหา Internet Server ได้เท่านั้น แล้วบันทึกสิ่งที่ทำและผลที่ได้
5. ลองคอนฟิก Firewall Policy ให้ Client สามารถ Browse WEB และ SSH ไปหา Internet Server ได้ แต่ห้าม PING แล้วบันทึกสิ่งที่ทำและผลที่ได้
6. (คำถาม) ถ้าต้องการกำหนดให้บาง IP จาก Inside สามารถ Access บาง IP ที่ Outside ต้องกำหนดอย่างไร
7. (คำถาม) ถ้าต้องการกำหนดให้บาง IP จาก Outside สามารถ Access บาง IP ที่ Inside ต้องกำหนดอย่างไร
8. (คำถาม) จาก Guideline ที่ว่าเราควรใช้ NAT สำหรับการทำ Routing เท่านั้นโดยไม่ควรใช้มันโดยหวังผลเรื่องของ Security นศ เห็นด้วยหรือไม่ อย่างไร
9. (คำถาม) นศ คิดว่าการทำ Access Control โดยอุปกรณ์ Firewall เหมือนหรือแตกต่างจากการทำโดย Router หรือไม่อย่างไร



Zone	IP Address
Management	10.30.6.0/24
Outside	203.0.113.0/24
Inside	192.168.1.0/24

